

OKMULGEE, OK, USA

WMO: 722164

Lat: 35.668N

Lon: 95.949W

Elev: 720

StdP: 14.32

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53949

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 15.0	(c) 19.0	(d) 4.2	(e) 7.0	(f) 18.9	(g) 9.1	(h) 9.0	(i) 25.5	(j) 24.5	(k) 53.5	(l) 22.6	(m) 53.5	(n) 6.8	(o) 10	(p) 0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	101.6	77.0	98.8	76.8	95.0	76.8	81.1	93.1	79.7	91.2	78.4	89.8	9.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	153.7	86.7	76.9	144.1	84.7	75.3	136.1	83.4	45.2	92.7	43.8	91.1	42.5	90.4	97.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.5	(b) 19.8	(c) 18.0	(e) 7.0	(f) 105.7	(g) 5.8	(h) 6.7	(i) 2.9	(j) 110.5	(k) -0.5	(l) 114.4	(m) -3.7	(n) 118.2	(o) -7.9	(p) 123.0		
(a) 22.5	(b) 19.8	(c) 18.0	(e) 6.2	(f) 83.1	(g) 5.3	(h) 4.8	(i) 2.4	(j) 86.6	(k) -0.8	(l) 89.4	(m) -3.7	(n) 92.0	(o) -7.6	(p) 95.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.8	40.6	43.5	53.7	61.7	69.1	78.3	82.0	81.6	74.1	62.1	52.0	41.8		
	DBStd	17.17	10.80	11.50	10.41	8.93	7.37	5.29	5.52	6.07	6.76	9.18	9.79	9.99		
	HDD50	1055	326	242	77	15	1	0	0	0	0	13	91	290		
	HDD65	3244	758	608	371	161	47	0	0	0	9	167	403	720		
	CDD50	5363	34	59	192	367	592	850	992	980	723	388	150	36		
	CDD65	2079	1	5	21	63	173	401	528	515	282	77	12	1		
	CDH74	23089	10	34	167	655	1524	4436	6467	6140	2800	746	105	5		
	CDH80	11192	0	4	32	176	462	2099	3542	3385	1260	218	14	0		
(14) Wind	WSAvg	7.7	8.0	8.5	9.5	9.7	8.1	7.6	6.0	6.0	6.2	7.2	8.0	7.6		
(15) Precipitation	PrecAvg	40.7	1.7	2.0	3.4	4.3	5.6	4.4	3.1	3.0	4.1	3.9	3.4	2.4		
	PrecMax	56.4	4.3	5.6	9.4	11.2	10.2	10.5	9.4	11.9	11.0	15.1	9.1	8.7		
	PrecMin	21.4	0.0	0.2	0.0	0.4	1.8	0.7	0.0	0.0	0.0	0.0	0.0	0.1		
	PrecStd	7.6	1.1	1.5	2.0	2.3	2.5	2.2	2.7	2.1	3.0	3.2	2.3	2.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.8	79.3	83.9	91.3	91.0	100.4	106.1	108.2	99.2	90.0	81.6	73.3		
		MCWB	57.6	63.5	66.1	73.0	73.9	79.1	76.9	76.6	74.2	73.1	65.2	61.4		
	2%	DB	69.8	72.8	79.0	84.0	88.1	96.9	100.3	102.0	94.1	85.8	77.1	68.4		
		MCWB	55.9	59.8	63.1	68.3	73.9	76.3	76.8	76.9	75.2	71.2	62.1	57.5		
	5%	DB	64.1	67.7	74.4	81.1	84.7	92.8	98.7	98.9	91.0	81.7	72.6	63.2		
		MCWB	54.3	57.8	61.3	66.1	71.8	76.7	77.0	76.8	74.2	67.2	61.5	54.9		
	10%	DB	59.3	63.0	71.5	77.7	82.0	90.5	94.7	94.9	88.0	78.5	68.3	57.9		
		MCWB	49.1	53.7	59.3	64.9	70.6	76.0	77.5	76.5	73.2	66.1	58.2	50.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.4	66.4	69.1	74.8	78.8	82.4	83.5	82.2	78.8	75.3	69.6	65.6		
		MCDB	67.2	74.3	79.4	83.9	87.0	93.4	95.0	95.1	89.4	84.9	76.9	72.0		
	2%	WB	59.9	63.5	66.5	71.9	76.3	80.4	81.3	80.4	77.3	73.5	66.4	61.5		
		MCDB	66.1	69.3	75.1	80.8	83.9	91.5	92.3	93.7	88.2	82.3	73.2	65.9		
	5%	WB	54.4	59.6	64.3	69.2	74.6	78.7	79.5	79.2	76.1	71.5	63.4	56.2		
		MCDB	62.6	65.6	71.3	77.2	81.7	89.3	90.3	92.1	86.5	77.5	69.3	60.4		
	10%	WB	50.0	54.2	61.8	67.1	72.8	77.4	78.4	78.0	74.8	69.0	60.0	51.9		
		MCDB	58.3	61.5	68.8	74.6	79.5	87.9	89.7	90.1	84.4	75.1	66.9	57.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.7	21.1	22.1	22.2	19.7	20.2	21.7	22.2	22.8	23.3	22.9	20.1		
		MCDBR	29.9	28.8	26.0	26.1	22.4	24.0	26.0	27.1	26.6	26.2	27.1	27.7		
		MCWBR	18.7	17.7	14.3	14.0	10.6	9.5	7.5	7.7	9.7	12.4	15.8	18.3		
		MCDBR	26.3	25.2	20.8	21.6	19.2	21.4	21.0	22.8	21.7	20.3	21.9	22.3		
	5% WB	MCWBR	18.9	18.1	12.8	13.9	10.6	10.2	8.2	8.7	8.8	10.8	16.2	18.2		
(40) Clear-Sky Solar Irradiance	taub		0.301	0.309	0.350	0.385	0.421	0.429	0.441	0.441	0.393	0.350	0.319	0.298		
	taud		2.496	2.492	2.407	2.322	2.285	2.292	2.303	2.263	2.368	2.477	2.507	2.517		
	Ebn at Noon		287	297	292	285	275	272	267	265	273	278	278	281		
	Edh at Noon		26	30	35	40	42	42	41	42	36	30	26	24		
(44) All-Sky Solar Radiation	RadAvg		825	1000	1321	1643	1799	2064	2095	1900	1580	1189	876	700		
	RadStd		85	149	83	105	171	214	121	121	137	157	109	88		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page